University of Western Australia

Honours Project

# Knowledge Graph Construction of Patient Operation Reports

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# Abstract

Clinical documentation in healthcare remains highly unstructured and inconsistent, posing challenges for automated analysis and interoperability, especially in Western Australia. These documents are difficult to analyse at scale due to challenges of computational readability, data fragmentation, and the resource requirements for processing and storage. This study investigates whether large language models (LLMs) can transform free-text surgical operation reports into structured, semantically meaningful knowledge graphs that support ICD-10-CM principal diagnosis classification. The prompting frameworks, KGGen, Extract–Define–Canonicalise (EDC), and Text2KGBench, were implemented using both general-purpose (Llama-3-8B, Llama-3-70B) and medically tuned (MedGemma-27B) models.

Performance was evaluated across multiple levels of ICD-10-CM granularity, revealing that while knowledge graph-based methods did not consistently outperform direct classification, they produced interpretable and standardised representations of operative information. A lightweight ontology was developed to capture surgical entities such as procedures, anatomy, findings, and personnel, providing a domain-specific schema suited to operative contexts. The results emphasise the potential of structured, ontology-guided representations to enhance interoperability and interpretability in clinical data and contribute to more patient-centric and semantically coherent healthcare information systems.

# 1 Introduction

## 1.1 Background and Motivation

The advancement of data digitisation introduced Electronic Health Records (EHRs), which have significantly changed how medical information is managed. EHRs offer a searchable and scalable way to store and retrieve patient data in structured form. They allow real-time updates, patient outcomes tracking, and seamless information exchange between providers. Structured data is organised into tables or predefined formats, helping to identify clear relationships between data points, such as a patient and their medical history. While structured data is common in healthcare, particularly through EHRs, it is often siloed within individual institutions and lacks interoperability across systems. This separation limits the full use of available information. Efforts such as FHIR (Fast Health Interoperability Resources) [1] and Australia's My Health Record [2] aim to address this issue by promoting standardised data exchange. However, unstructured clinical data makes up about 80% of EHRs [3]. These unstructured forms, such as PDFs, scanned documents, and isolated databases, further reduce interoperability between institutions [4]. Accessing this information is therefore important to create complete and accurate patient records. Advances in Natural Language Processing (NLP) and Large Language Models (LLMs) have made it increasingly possible to extract useful information from unstructured data sources [5], [6].

As a result, recent research continues to focus on using more meaningful and semantically rich representations to collate unstructured and structured EHR data of patient records [7], [8]. One such representation that has gained increasing interest in this context is knowledge graphs. Knowledge graphs represent information as a network of interconnected entities and relationships, where nodes denote concepts such as patients, procedures, or diagnoses, and edges describe how these concepts are related. This structure enables the integration of heterogeneous data sources by providing a unified, semantically consistent framework that connects structured database entries with information extracted from unstructured clinical text [9]. This not only enhances interoperability between disparate data types but

also supports downstream applications such as automated coding, clinical summarisation, and decision support [9].

Recent studies have demonstrated the growing role of knowledge graph construction (KGC) in healthcare, particularly in integrating diverse clinical data and improving explainability in medical AI systems. Abu-Salih et al. (2023) conducted a systematic review highlighting that healthcare KGs are increasingly being used to unify structured databases, ontologies, and unstructured text for tasks such as patient summarisation and diagnostic support [10]. Rajabi and Kafaie (2022) discussed how knowledge graphs contribute to explainable AI (XAI) by linking model outputs to interpretable, clinically relevant concepts [11]. In recent hybrid approaches, knowledge graphs have been integrated with symbolic reasoning to create interpretable AI systems capable of handling heterogeneous medical data sources [12]. Collectively, these works demonstrate that KGC is becoming a critical foundation for interoperable, transparent and context-aware AI solutions in modern healthcare, enabling consistent integration of structured and unstructured EHR data across institutions.

## 1.2 Research Aim and Project Focus

In response to the growing need for structured and semantically rich representations of unstructured EHR data, this project explores the use of large language models (LLMs) to automatically construct knowledge graphs (KGs) from unstructured surgical operation reports. Operation reports are a critical component of the clinical record, containing detailed procedural descriptions, findings, and post-operative plans. The findings document what the surgeon observed during the procedure, representing new clinical information that was not known beforehand, while the post-operative plan outlines the next steps in patient management. These reports, therefore, capture both the procedural context and the evolving state of the patient's condition, yet this information is rarely available in structured form. Beyond describing the surgical event itself, operation reports also carry implications for post-surgical care. They can indicate the expected duration of follow-up, suggest possible complications, and provide an implicit estimate of the patient's likely length of hospital stay. The lack of standardisation in how such reports are

written presents challenges for data reuse, clinical summarisation, and diagnostic classification. By transforming this free text into structured representations, it becomes possible to create interoperable and interpretable summaries that encapsulate the essential details of each surgical encounter.

## 1.3 Contribution

To achieve this, this research evaluates prompting-based knowledge graph construction frameworks [13]–[15] that guide LLMs through the process of extracting, defining, and linking clinically relevant entities and relationships. Three techniques are investigated: KGGen, which performs a two-step extraction and relation-linking process [13]; EDC (Extract–Define–Canonicalise), which formalises extraction utilising examples within the prompt [14]; and Text2KGBench, which applies an ontology-based prompting template aligned with the structure of operation reports [15]. Each framework is implemented and assessed across multiple LLM architectures, including general-purpose and medically tuned models, to examine how model scale and domain alignment influence the quality of extracted information. The constructed knowledge graphs are then applied to a downstream classification task, in which the goal is to predict the ICD-10-CM (International Classification of Diseases, 10th Revision, Clinical Modification) principal diagnosis code associated with each operation report. This task provides a quantitative basis for comparing the utility of different graph construction strategies, while also offering insight into how structured representations affect diagnostic reasoning.

This research aims to demonstrate a pathway toward patient-centric summaries that consolidate procedural and contextual details from unstructured text into semantically coherent, machine-interpretable forms. By leveraging LLMs within controlled prompting frameworks, the study seeks to bridge the gap between raw clinical narratives and structured electronic health records, contributing to the broader goal of improving interoperability, transparency, and the reusability of surgical documentation in modern healthcare systems.

## 1.4 Outline

- **Chapter 1: Introduction** outlines the motivation behind this research and the challenges associated with unstructured clinical documentation. It introduces the role of large language models (LLMs) in constructing knowledge graphs (KGs) for surgical operation reports and defines the key objectives of the study.
- **Chapter 2: Literature Review** examines prior research in knowledge graph representation and natural language processing (NLP) for healthcare. It discusses the evolution of encoder-based and LLM-based information extraction methods, as well as prompting frameworks such as KGGen [13], Extract–Define–Canonicalise (EDC) [14], and Text2KGBench [15] that inform the methodology of this work.
- **Chapter 3: Methodology** details the dataset from Perth Children's Hospital, including data preparation, alignment, and sampling. It explains the ontology design, prompting setup, and model configurations used for knowledge graph construction and downstream ICD-10-CM classification.
- **Chapter 4: Results** presents the findings of this research. It reports performance across different prompting frameworks and models, and includes visualised examples of constructed knowledge graphs to demonstrate variations in granularity and semantic coverage.
- **Chapter 5: Discussion** interprets the results in the context of clinical relevance. It analyses how model scale, ontology structure, and medical domain alignment influence performance, and discusses key limitations related to model outputs and evaluation design.
- **Chapter 6: Conclusion** summarises the main contributions of the research and reflects on its implications for interoperability and patient-centric clinical summarisation. It also identifies opportunities for future work, including integrating temporal reasoning and extending the framework toward longitudinal knowledge graph construction.

# 2 Literature Review

This chapter reviews the key areas of research that underpin this project. It begins by examining the role of knowledge graphs in structuring and representing clinical information, with particular focus on their application to medical data and interoperability. The following section explores natural language processing (NLP) techniques used in ICD coding, highlighting how encoder models capture semantic meaning from clinical text. Subsequent sections discuss the process of knowledge graph construction (KGC), including information extraction and recent prompting frameworks such as KGGen, EDC, and Text2KGBench [13]–[15]. Finally, the chapter outlines developments in leveraging large language models (LLMs) for ICD coding and the associated ethical and practical considerations when applying these methods in clinical settings.

## 2.1 Knowledge Graphs (KG)

Knowledge graphs are structures that represent data as a network of related entities. In this graph-based structure, nodes represent entities and edges describe the relationships between entities [16], [17]. The nature of these entities and relationships has context-specific semantics that allow users to understand the connections. The semantics are derived from pre-existing frameworks known as ontologies, which are formal representations of concepts within a specific domain, along with their properties and the relationships between them (Figure 1). Most constructed knowledge graphs do not have an explicit ontology. However, ontologies are useful for comparing graphs constructed for similar purposes and allow subgraphs to be linked under the same ontology.

In the medical domain, knowledge graphs have gained interest for their ability to structure heterogeneous clinical data and medical definitions [8]. By capturing relationships between clinical concepts in a structured and interpretable way, clinicians can trace how an outcome was made based on connected entities. Recent studies have also highlighted the role of knowledge graphs in enhancing Explainable AI (XAI) models in healthcare [8]. They facilitate the detection of healthcare

Figure 1: An example medical knowledge graph [10]

misinformation, adverse drug reactions, and drug-drug interactions, thereby reducing the knowledge gap between healthcare experts and AI-based models [11]. In this context, ensuring that AI systems provide explainable outputs is not just beneficial but increasingly important to ensure trust and clinical accountability [18].

Systematised Nomenclature of Medicine Clinical Terms (SNOMED-CT) [19] and the International Classification of Diseases (ICD) [20] are standardised medical terminologies popular across institutions. These vocabulary sets are widely used in clinical documentation and are regularly maintained to reflect advances in medical research and practice. ICD codes begin with a letter prefix followed by alphanumeric characters up to a length of 7. The length of a code reflects its specificity, which describes finer details such as anatomical location, cause and severity (Table 1).

Typical integration is achieved through manual coding; however, there have been investigations in automating this with Generative AI (GenAI) [21] and NLP techniques [22], [23]. By utilising automation, the potential arises for investigating

| Code | Code Description |
|---|---|
| **S52** | Fracture of forearm |
| **S52.5** | Fracture of lower end of radius |
| **S52.52** | Torus fracture of lower end of radius |
| **S52.521** | Torus fracture of lower end of right radius |
| **S52.521A** | Torus fracture of lower end of right radius, initial encounter |

Table 1: Example of increasing specificity in ICD-10 code structure

the alignment of entities and relationships with evolving ontologies in a way that better maintains semantic consistency and improves interoperability.

## 2.2 Natural Language Processing Techniques for ICD Coding

For ICD code prediction, encoder models serve an important role by transforming raw clinical text into representations that can be compared against known descriptions of ICD codes [23]. Since ICD codes are often inferred from descriptions of conditions or procedures, robust embeddings that capture semantic meaning are essential. For instance, similar procedures or diagnoses may be described using different wording across institutions or practitioners. High-quality vector embeddings allow for these language differences to be resolved in the embedding space, enabling models to identify the most appropriate ICD code even when the match is not obvious. The effectiveness of this process is closely tied to the quality of the encoders, making domain-specific models particularly valuable in encoding clinical notes for downstream classification tasks such as ICD coding.

Beyond developing robust embeddings, NLP-based approaches have demonstrated considerable success across diverse clinical departments in extracting structured information from unstructured text. Recent reviews highlight applications in internal medicine, oncology, radiology, pathology, and surgical domains, where NLP systems have achieved strong performance in identifying diagnoses, risk factors, and post-surgical complications [24]. For example, entity recognition and linking methods have been used to extract pathological findings, medication histories, and

procedural details from free-text reports, achieving accuracies above 90% in several hospital implementations [24].

### 2.2.1 Encoder Models

Encoder models represent words as short, dense vectors, known as embeddings. Vector embeddings are dense, low-dimensional representations (between 50 - 1000) of words that capture semantic relationships more effectively than traditional sparse vectors, which are high-dimensional and based on vocabulary size or document frequency [25]. These dense vectors not only reduce the complexity of models by limiting the number of parameters but also better capture similarities between words by placing them closer together in the embedding space [25]. Capturing the similarity between texts is done by generating these embeddings from word and sentence-level encoders.

#### 2.2.1.1 Word Encoders

Popular models for contextualised encoding include Bidirectional Encoder Representations from Transformers (BERT) and Embeddings from Language Models (ELMo). These models generate high-dimensional vector representations of words, capturing semantic and syntactic context. While both models have broad applications, BERT has demonstrated superior performance in the clinical and biomedical domain [26], [27]. A key advantage of BERT is its adaptability through fine-tuning, enabling models like ClinicalBERT [28], BioBERT, and Bluebert [27] to better capture domain-specific language. Further refinements, such as ClinicalBioBERT, which is initialised from BioBERT, illustrate the continued evolution of domain-specific encoders [29]. The motivation for fine-tuning encoder models is to improve the representation of terms specific to a context. General-purpose embeddings often fail to represent specialised medical vocabulary accurately, as they are not trained on such data [30]. This limitation can be thought of as asking a toddler to spell complex words without seeing them before.

#### 2.2.1.2 Sentence Encoders

Sentence encoders are able to construct sentence vector representations by avoiding pooling techniques. An example is the Universal Sentence Encoder (USE), which utilises deep averaging networks (DAN) or transformer-based encoders to generate fixed-length embeddings for entire sentences [31]. These approaches present positive results to produce sentence embeddings for classification and semantic similarity [31]. However, similar to general word-based encoders, USE and general sentence encoders fall short of word-based encoders that use mean pooling to capture sentence-level semantics [32]. This is especially important when comparing the similarity between two sentences because the semantics of each sentence are what governs its meaning.

## 2.3 Knowledge Graph Construction (KGC)

To construct knowledge graphs from clinical text, it is first necessary to extract meaningful entities and the relationships between them. This process is known as Information Extraction (IE), and it forms the backbone of knowledge graph generation in healthcare settings. IE enables unstructured clinical narratives, such as discharge summaries and progress notes, to be transformed into structured representations of medical knowledge. With advancements in language models, both traditional transformer-based architectures and large language models (LLMs) have been leveraged for this task. The following section outlines the key components of IE, including Named Entity Recognition (NER) and Relation Extraction (RE), and explores the evolving role of LLMs in improving the accuracy and scalability of clinical knowledge graph construction.

### 2.3.1 Information Extraction (IE)

Knowledge graph construction falls under the broader task of information extraction, which primarily involves Named Entity Recognition (NER) and Relation Extraction (RE). NER is responsible for identifying key entities within unstructured text, while RE determines the semantic relationships between these entities, both of which are fundamental to the process of constructing knowledge graphs. NER

plays a critical role in extracting essential information such as diseases, medications, and procedures, enabling their representation in a structured, summarised format. To perform these tasks, transformer-based models such as those built on BERT architectures are commonly used (Section 2.2.1). This process entails challenges due to the complexity and variability in medical language. As a result, clinically and medically trained models are preferred as they consistently outperform general-purpose ones in NER tasks [33]. This performance gain also extends to RE, particularly in clinical and biomedical domains [34].

#### 2.3.1.1 Large Language Models for Information Extraction

Recent research has also explored LLM-driven KG construction, where large language models are prompted to generate structured triples, which are sets of three elements that represent a factual statement in the form (subject, relation, object), directly from text [13], [15]. Frameworks such as KGGen [13], EDC [14], and Text2KGBench [15] have been developed to investigate different prompting strategies for guiding LLMs through this extraction process. The LLM approach continues to perform NER and RE by constructing specific prompts to do this [13]. Since we are using similar steps, why should we use an LLM over other language models? An advantage of LLMs is their ability to leverage broader contexts due to their larger context windows and more diverse training. This is increasingly beneficial in the clinical domain as entities and their relationships are often context-dependent and spread across multiple sentences or entire texts. In contrast, traditional models such as BERT are generally restricted to shorter sequences and may require adjustments to handle longer inputs. Although LLMs are capable of processing longer sequences of text, they are subject to limitations such as the Lost in the Middle phenomenon. This issue refers to the model's tendency to overlook information that appears in the middle of a long context as input. Liu et al. (2023) demonstrated that model performance is often highest when key information is located near the beginning or end of the input sequence [35].

This presents a significant challenge in domains like healthcare, where critical information, such as diagnosis summaries, treatment decisions, or surgical outcomes, may be embedded anywhere within a clinical note. In particular, clinical docu-

mentation often follows a progressive, chronological format that evolves throughout a patient's hospital stay. These notes can be lengthy and cumulative, with early entries capturing symptoms and test results, and final entries reflecting updated results or discharge planning. Consequently, important facts may not appear at optimal positions within the text, making models vulnerable to missing critical details. A straightforward workaround involves querying an LLM multiple times using segmented chunks of the original text. However, this approach risks overlooking important semantic relationships between entities that span across different chunks. This issue can be minimised by introducing overlap between adjacent chunks, ensuring that both preceding and succeeding contexts are preserved.

### 2.3.2 Knowledge Graph Assessment

Assessing the quality of knowledge graphs remains an important task in their construction and application. To introduce knowledge graphs as a method to summarise patient records, there needs to be an evaluation framework to understand how they represent and compile discharge summaries, clinical notes and patient records. This evaluation can be split into two separate yet interrelated metrics: intrinsic and extrinsic [36]. Intrinsic metrics aim to enhance performance on extrinsic tasks, while the effectiveness of an intrinsic metric is often reflected in its ability to support strong extrinsic outcomes [36].

Extrinsic evaluation assesses the knowledge graph in downstream applications. The quality is therefore measured by its impact on tasks such as question answering or clinical outcome prediction by comparing to ground-truth[1] predictions with appropriate classification metrics such as precision, recall and F1-score. Intrinsic evaluation focuses on assessing the structure and semantics of the knowledge graph itself. This includes metrics such as accuracy (the degree to which the KG represents the original data) and completeness (whether the KG captures all relevant information) [36]. Completeness, for instance, can be evaluated by comparing the ICD codes present in a KG against a ground-truth set derived from corresponding clinical texts. In sensitive domains like healthcare, expert validation is often required to ensure clinical accuracy. However, clinician time is both limited and

[1] Outcomes/results that are known to be true

costly, making large-scale manual assessment of knowledge graphs impractical. This also complicates the comparison of different KG construction techniques. To address this, the Measure of Information in Nodes and Edges (MINE) was introduced as an automated evaluation metric [13]. MINE extracts fifteen key facts from the source text using an LLM, then evaluates whether each fact is represented in the KG, with an LLM serving as the judge [13]. By reducing dependence on domain experts, MINE enables more efficient and scalable comparisons of knowledge graph extraction methods.

#### 2.3.2.1 Clinical Evaluation

Following the evaluation process that Mo et al. (2025) developed, we can shift the context to focus on the medical and clinical domain. One practical strategy leverages available structured data, such as ICD codes, age, and sex, recorded by clinical coders during hospital admissions. These data points are a basic reference set of known facts. Similar to the MINE [13] approach, an evaluation pipeline can be implemented where an LLM is prompted to judge whether these known attributes of an individual patient can be accurately derived from the constructed knowledge graph. This provides a scalable, semi-automated way to assess how well the KG preserves clinical information, using real-world ground-truth data without requiring complete annotated datasets. Such an approach enables basic yet meaningful validation of KG quality, especially in healthcare settings where certain facts are consistently recorded across patient records.

## 2.4 Leveraging Large Language Models for ICD Coding

This section introduces MedCodER, a framework that investigates ICD coding by combining information extraction and retrieval techniques powered by GPT-4 [21]. Its architecture supports flexibility across clinical domains while reducing dependence on annotated data. Beyond annotation, the framework holds potential for integrating ICD codes into knowledge graphs, enhancing semantic interoperability even when such codes are not explicitly stated in the text. The following subsection highlights critical concerns around the use of cloud-based LLMs in healthcare,

particularly regarding data privacy and regulatory compliance.

### 2.4.1 MedCodER

Initial ICD coding investigations found that out-of-the-box LLMs perform poorly on medical coding tasks [37], [38]. The MedCodER framework reformulates the ICD coding task as a combination of information extraction and retrieval using OpenAI's GPT-4 model [21]. This enables the model to map clinical narratives to ICD codes without requiring fine-tuned supervision. Specifically, it leverages zero-shot[2] prompting to query the model for relevant diagnoses or procedural codes directly from clinical text. This architecture allows the system to remain flexible and generalise across clinical domains while minimising reliance on labelled training data.

The ultimate aim of Baksi et al. (2024) was to design a framework that could serve as an AI assistant for clinicians responsible for medical coding [21]. Rather than using it solely for automatic text annotation, can the framework also be leveraged to integrate ICD codes into knowledge graphs? By embedding them within an ontology structure, even when those codes are not explicitly mentioned in the clinical text, the gap in structure and style between institutions can be reduced and increase interoperability.

### 2.4.2 Concerns with Cloud-Based LLMs

Continuous advancements in generative AI models, such as OpenAI's GPT, have led to their widespread adoption and application across a variety of areas. However, when deploying these models in sensitive domains like medicine, greater caution is warranted due to concerns around accuracy, reliability [39], and the potential consequences of erroneous outputs. An even worrisome concern arises within privacy and confidentiality when working with patient data on cloud-based models. An article on OpenAI's website explains that the models are trained on data from three main sources, one of which includes user-provided and user-generated content [40]. As a result, using cloud-based models with private data

[2]Zero-shot prompting is a technique that aims to instruct LLMs to perform specific tasks that it was not trained for

poses potential risks to confidentiality. This issue is especially significant in Australia as legislation, such as the My Health Records Act 2012, mandates that all patient data, including backups, must remain within Australian borders. Transmitting data overseas, even unintentionally, may breach national data sovereignty laws and trigger substantial penalties. Fortunately, local deployments of LLMs are available, and many API-accessible models, such as Large Language Model Meta AI (LLaMA), also have local equivalents. This enables sensitive data to be processed securely, thereby mitigating concerns around privacy and data leakage, which becomes particularly relevant where patient records must be handled with strict confidentiality.

## 2.5 Summary

The techniques discussed provide great interest in building on extracting meaningful patient knowledge graphs. By leveraging domain-specific encoders, structured ontologies, and LLM-driven information extraction, these methods enable the transformation of complex clinical narratives into structured, interpretable representations. As the field advances, ongoing challenges remain, particularly around maintaining up-to-date medical knowledge, handling fragmented or siloed data. Continuing on advancements in ICD automation, knowledge graphs can be constructed alongside information representations that support regular updates and enhance interoperability across healthcare systems. Furthermore, NLP techniques and generative AI open promising avenues for scalable, transparent, and clinically useful applications. Future work will focus on improving longitudinal KG maintenance (changes in patient records), refining evaluation metrics, and enhancing interoperability across healthcare systems through standardised, semantically enriched representations.

# 3 Methods

Clinical documentation within healthcare systems remains highly heterogeneous, particularly in operative settings where reports are composed as lengthy, unstructured narratives. These inconsistencies in style, terminology, and formatting pose challenges for both automated analysis and interoperability across health information systems. The problem addressed in this research is the lack of a standardised, structured representation of operative information that can support consistent and interoperable downstream tasks such as ICD-10-CM principal diagnosis coding. To address this, the study investigates whether large language models (LLMs) can be guided to transform free-text operation reports into structured, knowledge graph representations that capture clinically meaningful entities and relations. An overview of this methodological workflow is presented in Figure 2, which illustrates the progression from raw clinical text through knowledge graph construction to diagnostic classification. By examining prompting strategies such as KGGen, EDC, and Text2KGBench, this research seeks to develop a method for representing surgical narratives in a semantically coherent and machine-interpretable form, thereby advancing the standardisation and interoperability of clinical information within and across healthcare institutions.

## 3.1 Data

The sensitive clinical data investigated in this research are from the Anaesthesia and Pain Department of Perth Children's Hospital (PCH) from the ALLIGATOR database. The dataset is comprised of two related components, the first containing the clinical text regarding a patient's hospital visit and the second outlining the ICD codes that have been assigned to that visit. Approval for the use of this data was obtained from both the University of Western Australia (UWA) and the Western Australian Department of Health (WA Health) Human Research Ethics Committees.

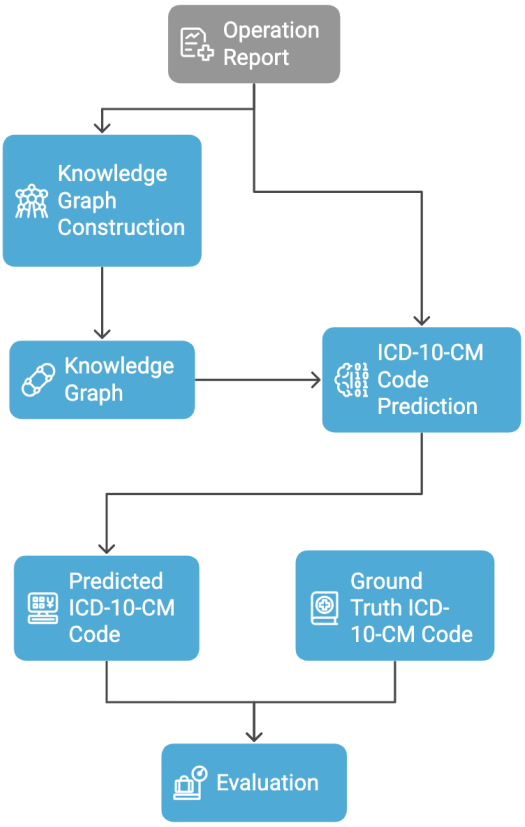


Figure 2: Overview of the experimental workflow illustrating the sequential stages of knowledge graph construction and principal diagnosis classification using large language models.

### 3.1.1 Clinical Text Datatset

This dataset lists various clinical files that have been extracted from their original PDF files. These file types include admission forms, progress notes, discharge summaries and operation reports. This project focuses solely on the operation reports, as they provide direct procedural context relevant to the downstream task of ICD coding. Furthermore, the operation reports typically contained a similar structured (Figure 3) presentation across other reports.

Each row in this dataset corresponds to a single document and includes patient-related identifiers such as name, date of birth, the Unique Medical Record Number (UMRN) and an Episode ID, which denotes a specific hospital visit.

**Operation Report**

(Patient and Surgeon Information)

---

**Operation and Principal Diagnosis**

(Diagnosis, Findings, Procedure)

---

**Authorisation**

(Signature / Date)

Figure 3: Example structure of an operation report in the dataset

### 3.1.2 ICD Code Dataset

This dataset contains every ICD-10-CM (Clinical Modification) code related to a patient's hospital visit, which has been done by PCH's in-house clinical coders. Similar to the Text dataset, this data contains the specific patient's UMRN and episode identifier.

For a given hospital visit, multiple ICD-10-CM codes may be assigned as a patient may come in with various complications or secondary conditions alongside their principal diagnosis. As a result, for each surgery, there is a singular principal diagnosis including ICD-10-CM codes related to other conditions, external causes, activity, place of occurrence and morphology. The dataset distinguishes these categories through the "Coding Type" field, which separates principal and secondary diagnoses as well as other relevant classifications.

| **Columns** | **Description** |
|---|---|
| UMRN | Unique identifier for each patient |
| Surgery Date | Date of the surgical procedure (YYYY-MM-DD) |
| Episode ID | Specific patient visit |
| Coding Type | Specifies the role of the ICD-10-CM code |
| Diagnosis Code | Assigned ICD-10-CM code |
| Diagnosis | Description of assigned ICD-10-CM code |

Table 2: Structure of ICD-10-CM Dataset

### 3.1.3 Data Alignment

Ideally, the two datasets would be related through a shared identifier, such as the combination of UMRN and Episode ID. However, because the Episode IDs are recorded differently and independently in each dataset, this approach is not feasible. The reason for this is that Episode IDs can change over the course of an admission, for example, emergency, inpatient and or outpatient. As a result, they aren't always linked to the Episode ID in the ICD dataset. The alternative is to use the surgery date in combination with the UMRN to establish a match. While the ICD dataset explicitly records the surgery date, the text dataset contains only a general date field that reflects when the file was saved or uploaded by the clinician, rather than the actual date of surgery. This introduces complications, as the saved date may not coincide with the operation date, making direct matching unreliable. Instead, the surgery date is extracted from within the operation reports themselves using regular expressions, providing a more accurate basis for alignment with the ICD dataset.

### 3.1.4 Target Dataset

Before performing any matching, the clinical text dataset contained 11,247 operation reports corresponding to 8,201 unique patients, while the ICD dataset included records for 8,330 unique patients. Across both datasets, 8,106 UMRNs were common. After this filtering step, 6,031 operation reports remained with corresponding principal diagnoses listed in the ICD dataset. To reduce runtime while preserving diagnostic diversity, we further stratified the dataset based on the *initial ICD-10-CM category* of the principal diagnosis (the first character of the code). Each initial category was treated as a stratum (with a maximum of 26, corresponding to the lexical value), ensuring that sampling reflected the underlying distribution of diagnostic groups. From the 6,031 matched operation reports, we drew a stratified sample of 1,000 cases.

This proportional stratification aims to maintain representation across common and rare diagnoses, while limiting the overall dataset size to one that is computationally manageable for repeated experimental runs. Sampling was performed

without replacement using a fixed random seed, guaranteeing reproducibility and comparability across experiments.

The resulting dataset, therefore, balanced two objectives:

1. Retaining broad coverage of ICD-10-CM categories to make evaluation meaningful
2. Reducing computational costs associated with multiple runs of large language models during knowledge graph construction and classification.

The resulting target dataset is summarised in Table 3, which outlines the key columns and their descriptions.

| **Columns** | **Description** |
|---|---|
| UMRN | Unique identifier for each patient |
| Name | Name of the patient |
| Date of Birth | Patient's date of birth |
| Episode ID | Specific patient visit |
| Surgery Date | Date of the surgical procedure (YYYY-MM-DD) |
| Principal Diagnosis | Assigned ICD-10-CM code as the Principal Diagnosis |
| Text | Procedure related part of the operation report (Figure 3) |
| Full Text | The entire operation report from the Clinical Text dataset |

Table 3: Structure of Target Dataset

## 3.2 Knowledge Graph Construction (KGC)

This project examines the use of prior methods KGGen (Knowledge Graph Generation), EDC (Extract, Define, Canonicalise), and Text2KGBench (Text to Knowledge Graph Benchmark) as prompting strategies for knowledge graph construction [13]–[15]. These approaches employ different prompting designs: zero-shot prompting, few-shot prompting, and few-shot prompting augmented with an ontology structure. By implementing and comparing these strategies, this work aims to identify which prompting style yields more effective and accurate information extraction based on the classification of the principal diagnosis.

To operationalise these prompting strategies, the models were loaded via the Hugging Face Transformers library and executed using the `pipeline` function for text generation. Each operation report was passed as input to the pipeline together with the designed prompts (corresponding to KGGen, EDC, or Text2KGBench). In all cases, the extraction instructions (prompt templates) were provided as *system messages*, while the unstructured operation report text was supplied as the *user message*. This ensured a clear separation between task definition and content input, thereby enforcing consistent behaviour across runs. The pipeline standardises the model interaction by automatically handling tokenisation, batching, and text decoding, which ensures reproducibility across experiments.

When passing the operation report to the LLM in the *user message*, we pass a flattened version of the original text. By flattened, we mean that instead of providing the report in its native structure (section breaks and line spacing, Appendix E), we collapse the content into a single, continuous string. The reason for this flattening is to remove formatting irregularities (such as blank lines, inconsistent spacing, or alignment artifacts) and to normalise the input into a format that the model can process more efficiently (Figure 4). Flattening also reduces the number of tokens consumed by whitespace and formatting, thereby maximising the effective use of the context window.

```
":** Excision of midline dermoid cyst- glabllar region ####
**FINDINGS:**  1cm rounded cyst tightly attachewd to
underlying periosteum ## **PROCEDURE DETAILS:** GA,
LA, IVABs Vertical midline incision Cyst excised completely
with underlying attached periosteum Sent for histopathology
Layered closure with 5-0 Monocryl Skin 5-0 Monocryl pull
through monocryl Steristrips, Fixomull ## **LOCAL
ANAESTHETIC:** Yes 3ml .2% Ropivacaine + Adrenaline
**SURGICAL COUNT CORRECT:** Yes ##
**POSTOPERATIVE INSTRUCTIONS:** RPAO Discharge
when safe on simple analgesics Keep dressing dry 48 hrs the
allowed to wet and pat dry PDC review in 10-14 days ####
**DRAINS:** No **"
```

Figure 4: Example flattenning of operation report

### 3.2.1 KGGen

This prompting technique structures its prompts similarly to conventional information extraction tasks. That is, it separates the process into two stages, the first being named entity recognition and the second being relation extraction (Appendix B) [13]. These tasks are presented to the model in a stepwise manner, encouraging it to first identify relevant entities within the unstructured input text before progressing to the extraction of the semantic relations that link them. Framing entity recognition and relation extraction as sequential subtasks provides a structured procedure for knowledge graph generation, helping to ensure that each step contributes distinctly to the formation of coherent graph elements.

**Implementation:**

1. **Named Entity Recognition (NER)**: The LLM is first prompted with the initial KGGen template (Appendix B). This template is passed as a *system message* to define the extraction task, while the operation report is provided as the *user message*.

2. **Entity Preparation**: The recognised entities from step one were extracted and passed into the next stage of the pipeline.

3. **Relation Extraction (RE)**: A second KGGen prompt, again provided as a *system message*, was provided to the LLM with the extracted entities passed in as well. Similar to step one, the operation report is passed through the *user message*, allowing the model to identify and characterise semantic relations between the entities.

### 3.2.2 EDC

The Extract–Define–Canonicalise (EDC) prompting strategy is a knowledge graph construction (KGC) technique designed to guide large language models through a structured process of transforming text into triples. The framework divides the task into three distinct stages (extract, define, canonicalise) [14]; however, in this project, only the first stage was implemented. The focus remained on identifying relevant entities and relations rather than on downstream refinement or ontology

alignment.

1. **Extract**: The LLM is prompted to identify relevant entities and relations directly from the input text, producing raw candidate triples. This stage prioritises broad coverage, aiming to capture as much potentially relevant information as possible (Appendix C).

2. **Define**: In a full EDC workflow, the extracted triples would be refined by aligning entities and relations with human-readable descriptions or contextual qualifiers.

3. **Canonicalise**: The final stage involves normalising the defined entities and relations against an external ontology or controlled vocabulary.

By limiting the scope to the extraction phase, this study isolates and evaluates the core information retrieval capability of the prompting framework, providing a clear and controlled assessment of how effectively large language models can identify clinically meaningful triples from unstructured operative text. The implementation is intentionally left at this stage to serve as an example of a few-shot extraction prompt.

The implementation followed the same procedure as described for KGGen (Section 3.2.1) with one prompt only, as EDC skips the intermediate decomposition into separate named entity recognition and relation extraction stages. Instead, the EDC prompt directly instructed the model to extract triples in a single step. The EDC extraction prompt was provided as a *system message* to specify the task at hand, while the clinical text of the operation report was supplied as the *user message*.

### 3.2.3 Text2KGBench

Text2KGBench is a prompting framework originally designed to benchmark the ability of large language models to perform knowledge graph construction (KGC) across different domains [15]. Text2KGBench provides a direct prompting strategy where the model is asked to extract triples from text in a single task. Importantly, this approach supports adherence to an explicit ontology structure, guiding the

model to align extracted entities and relations with predefined node and edge types (Appendix D).

The framework encourages models to:

1. **Extract entities and relations**: Identify relevant elements in the input text and express them as head–relation–tail triples in this format relation(head, tail).
2. **Align to ontology schema**: Ensure that extracted triples adhere to the defined entity and relation types.

In this project, we implemented the extraction step with ontology adherence by constraining outputs to our defined property-graph schema (see Section 3.2.4). This allowed us to evaluate whether explicitly embedding ontology structure in the prompt improved the quality of extracted knowledge graphs and their downstream utility for principal diagnosis classification.

The implementation mirrored the procedure used for KGGen and EDC. The Text2KGBench-style prompt was provided as a *system message* to specify the extraction task and enforce adherence to the ontology, while the operation report text was supplied as the *user message*.

### 3.2.4 Ontology Design

The ontology developed for this research was designed to capture the core semantic structure of surgical operation reports, focusing on the relationships that characterise the procedural context, the patient, and the clinical entities involved. As shown in Figure 5, the ontology is centred around the patient node, which acts as the anchor for connecting both procedural and contextual information.

Entities such as procedure, surgeon, anatomy, anaesthesia, medication, and findings represent the major conceptual components commonly described within operative narratives. Each node type corresponds to information explicitly documented in operation reports. For example, the procedure node captures the operation performed, while anatomy denotes the anatomical site or system on which the operation is carried out. The surgeon entity links via the performs relation

to the procedure, reflecting operator involvement, and findings capture operative outcomes through the resulted_in relation.

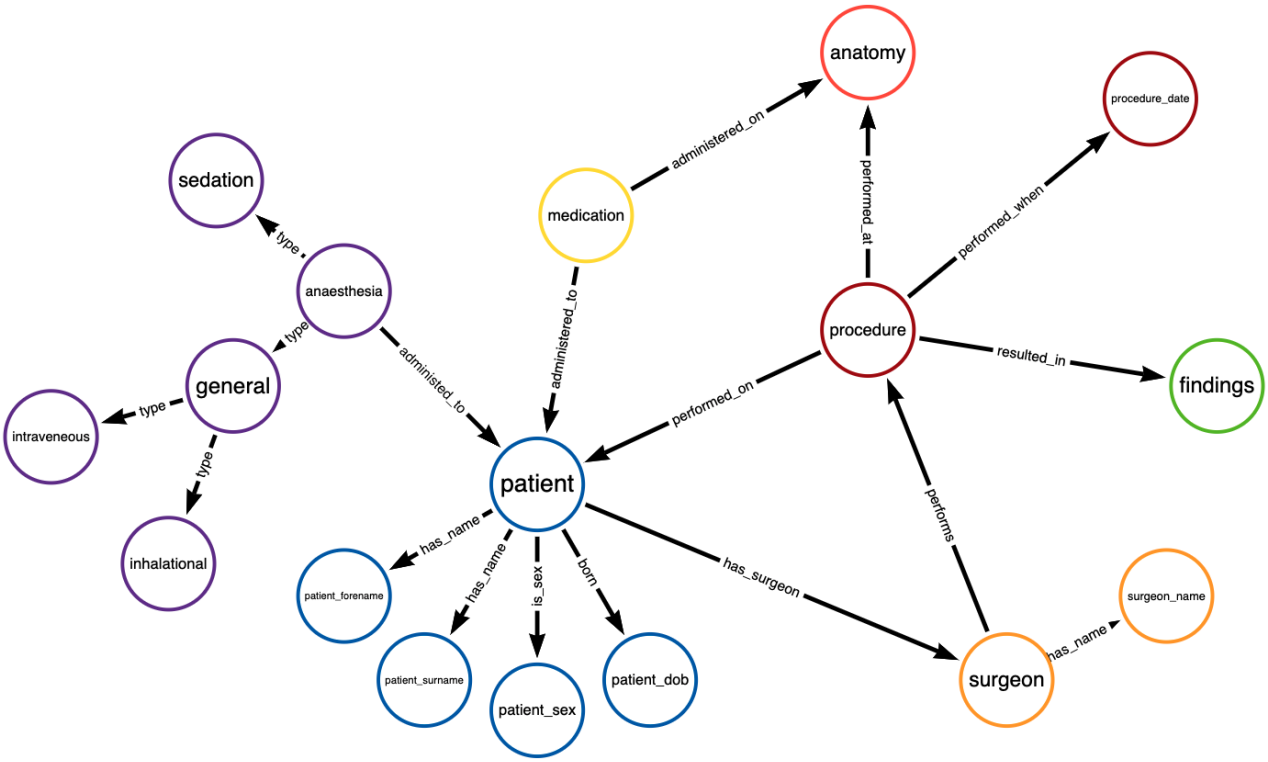


Figure 5: Visualisation of the designed ontology for knowledge graph construction using Text2KGBench

The structure more closely resembles a lightweight schema designed for targeted knowledge graph construction rather than a fully axiomatic ontology such as SNOMED-CT [19]. SNOMED-CT provides an extensive hierarchical model of clinical concepts and their logical relationships, supporting formal reasoning and comprehensive interoperability across healthcare systems. In contrast, the ontology developed in this research is deliberately simplified and task-specific, focusing only on the entities and relations most relevant to operative reporting. It instead prioritises interpretability, compactness, and practical extraction from natural language text. This streamlined design allows it to function effectively within the limited scope of surgical documentation, while still maintaining conceptual com-

patibility with larger clinical terminologies should future integration be required.

The ontology also encodes patient-specific attributes (e.g., sex, date of birth, forename, surname), allowing the representation of personalised information. Connections such as administered_to and performed_on further ground the relationships between patient, medication, anaesthesia, and procedure. This structure attempts to mirror the relational language of clinical documentation, where descriptions typically follow a sequence of who, what, where, and result.

Overall, this ontology provides a structured schema that constrains Text2KGBench's extraction process, ensuring that generated triples adhere to clinically meaningful entity and relation types. By reflecting the dependencies within operation reports, it supports the generation of coherent, interpretable knowledge graphs aligned with that of clinical reporting.

### 3.2.5 Large Language Models Investigated for Knowledge Graph Construction

The selection of large language models (LLMs) for this study is motivated by their potential to support automated knowledge graph construction (KGC) from text. Recent advances in LLM development have produced models with substantial variation in scale, training data composition, and domain specialisation, each of which may influence their effectiveness in extracting structured entities and relations. In this research, four representative models are considered: two general-purpose LLMs from the Llama 3 family (Llama-3.1-8B and Llama-3.3-70B), a domain-adapted medical model (MedGemma-27B-Text-IT), and a smaller instruction-tuned model optimised for efficiency (Mistral-Small-Instruct-2409) (Table 4). Together, these models provide a basis for examining how differences in model architecture and training objectives affect the capacity of LLMs to generate coherent and semantically consistent knowledge graphs within the clinical context.

#### 3.2.5.1 Meta-Llama

For this study, two models from Meta's Llama family were selected: Llama 3.1-8B and Llama 3.3-70B. The 8B parameter model represents a lightweight con-

| Model | Parameters | Tuning | Context Length | Release Date |
|---|---|---|---|---|
| Llama 3.1-8B | 8B | Instruction-tuned | 128K | July 2024 |
| Llama 3.3-70B | 70B | Instruction-tuned (GQA) | 128K | December 2024 |
| Mistral-Small-Instruct-2409 | 22B | Instruction-tuned | 32K | September 2024 |
| MedGemma-27B-text-it | 27B | Instruction-tuned (medical domain) | 128K | May 2025 |

Table 4: Key specifications of selected models

figuration optimised for efficiency, with reduced computational and memory requirements while still supporting long context windows (up to 128K tokens) and providing instruction-tuned capabilities suitable for text generation and extraction tasks. By contrast, the 70B parameter model in the newer Llama 3.3 release offers substantially greater reasoning capacity and improved alignment, benefiting from architectural enhancements such as grouped-query attention and updated instruction tuning. This model has been shown to perform comparably to much larger systems, making it a strong candidate for complex tasks without incurring the full computational overhead of ultra-large models. Including both the 8B and 70B variants allowed for an evaluation of performance across scales: the smaller model illustrates the feasibility of deploying resource-constrained systems, while the larger model serves as a more capable baseline that better reflects state-of-the-art instruction following and reasoning. Together, they provide a balanced comparison between efficiency and capability, aligning the evaluation with practical deployment considerations and the need for robust performance in knowledge graph construction and downstream tasks.

#### 3.2.5.2 MistralAI

We also include Mistral-Small-Instruct-2409, a 22 billion-parameter instruction-tuned model developed by Mistral AI (released in September 2024) supporting a context window of up to 32,000 tokens. Since it is more compact than many larger models but still capable of instruction following and reasoning, it serves as a

mid-scale alternative: more capable than lightweight models but less demanding in computation than large models. Its inclusion allows us to examine how a mid-sized, efficient, instruction-tuned model performs relative to both the small (e.g. 8B) and larger (e.g. 70B) alternatives, helping us explore trade-offs between resource usage and reasoning.

#### 3.2.5.3 MedGemma

In addition to the Llama models and MistralAI model, we also include MedGemma-27B-text-it, a specialist medical language model developed by Google under the Health AI Developer Foundations framework [41]. MedGemma is built on the Gemma 3 architecture and is instruction-tuned to perform strongly on medical reasoning, clinical question answering, and text generation tasks in the healthcare domain. This variant is trained exclusively on medical text and is optimised for inference efficiency while retaining very large context support. In benchmark evaluations, MedGemma-27B yields high accuracy on clinical QA tasks, outperforming untuned baseline models of a similar scale [41]. By including MedGemma alongside Llama models, we can assess how a domain-specialist model optimised for the medical domain compares to general-purpose models of differing scales; it serves as a strong domain-adapted baseline and helps clarify tradeoffs between domain specificity and general reasoning capability.

## 3.3 Principal Diagnosis Classification

### 3.3.1 Large Language Models Utilised for Principal Diagnosis Classification

For the task of principal diagnosis classification, two large language models were selected from the broader set evaluated in the knowledge graph construction experiments. The choice of models reflects an interest in comparing the performance of a large, general-purpose architecture (Llama-3-70B) with that of a smaller, domain-adapted medical model (MedGemma-27B). The same prompting and evaluation pipeline was applied across both models to ensure comparability, to assess their capacity to generate accurate ICD-10-CM principal diagnosis codes directly from

unstructured clinical text. In each case, the models are explicitly instructed to output the singular most appropriate ICD-10-CM code describing the principal diagnosis (Figure 6, 7), thereby standardising the expected response format and aligning it with clinical coding practice.

### 3.3.2 Classification from Plaintext (Baseline)

To establish a baseline, the operation reports were passed to the LLM in their flattened form using the prompt shown in Figure 6. In this representation, the text is collapsed into a single continuous line, removing all layout-based formatting such as line breaks and spacing irregularities.

This configuration tests the model's capacity to interpret the document when deprived of visual or positional structure, relying solely on lexical and syntactic information available within the flattened text. It therefore provides a conservative baseline for comparison: any subsequent gains observed when using structured inputs or knowledge-graph–based representations can be attributed to the additional contextual and relational information introduced by those approaches.

> You are a clinical coding assistant. Your task is to read a clinical note and return the one most appropriate ICD-10 code based on the content to describe the Principal Diagnosis. Use the official ICD-10-CM classification system.

Figure 6: Prompt to Classify the Principal Diagnosis from Plaintext

### 3.3.3 Classification from Constructed Knowledge Graphs

Prompting the LLM to classify the principal diagnosis from the constructed knowledge triples/graphs, from each pair of prompting and model, follows the same overall design as the baseline. The only modification was the input format, which was reformulated from "a clinical note" to “a list of related triples from a clinical note” (Figure 7). Maintaining consistency in prompt design across both settings reduces the number of confounding variables and ensures that any observed differences in performance can be attributed to the input representation rather than

to changes in prompt structure.

This control is essential for establishing a fair comparison between unstructured and structured inputs. By keeping the prompting framework constant, we can directly assess the extent to which knowledge graph construction contributes to classification performance, independent of alternative explanations such as prompt wording or formatting differences.

> You are a clinical coding assistant. Your task is to read a list of related triples from a clinical note and return the one most appropriate ICD-10 code based on the content to describe the Principal Diagnosis. Use the official ICD-10-CM classification system.

Figure 7: Prompt to classify the principal diagnosis from constructed knowledge graphs

#### 3.3.4 Extraction of Principal Diagnosis

To extract the principal diagnosis code from the LLM responses, we leveraged the fact that ICD-10-CM codes follow a well-defined structural pattern, beginning with a single letter followed by two digits, with optional alphanumeric extensions (Table 1). This predictable structure enabled the use of regular expressions to systematically capture valid codes within the free-text responses generated by the models.

This step was necessary because, despite explicit prompting in Figure 6, 7, LLMs did not always adhere to the instruction (Appendix F). In some cases, responses contained multiple candidate codes, explanatory text, or narrative justifications around the coding choice. Without post-processing, such variability would have made direct comparison across models unreliable.

To enforce consistency, only the first valid ICD-10-CM code detected by the regular expression was selected as the model's prediction. While this approach risks discarding potentially correct codes mentioned later in the response, it ensures that

each prediction corresponds to a single candidate diagnosis, avoiding subjective selection, and prevents inflation of accuracy metrics through multiple guesses.

Thus, regular expression matching served three key roles:

1. **Filtering for validity**: ensuring that only sequences conforming to the ICD-10-CM code structure were considered.
2. **Constraining output**: enforcing the one most appropriate choice by always selecting the first valid code.
3. **Standardising predictions**: guaranteeing comparability of outputs across different models and runs.

This method provided a robust and automated pipeline for converting noisy LLM responses into a consistent and evaluable coding format, which was essential for the downstream calculation of metrics.

## 3.4 Target Metrics

The evaluation of model performance is grounded in the ICD-10-CM coding framework, with the principal diagnosis serving as the reference point. Since the models were instructed to output the most appropriate ICD-10-CM code describing the principal diagnosis, the target metrics reflect both strict exactness and clinically meaningful closeness to the ground truth.

First, Exact Match Accuracy was measured by assessing whether the model's predicted ICD-10-CM code matched the annotated gold-standard code character-for-character. This provides a high-precision metric aligned with the expectations of clinical coding practice.

Second, we considered Category Match Accuracy, where predictions were deemed correct if the three-character category (e.g., S52 – Fracture of Forearm) matched the ground truth, even if further sub-classification (e.g., S52.52 vs S52.53) differed (Table 1). This reflects the fact that misclassifications often occur at the level of clinical modifiers (laterality, acuity, complication status), and a category-level agreement still indicates that the model has identified the appropriate diagnostic

domain.

Finally, we evaluated Initial Character Match, corresponding to the ICD-10-CM Initial Character level (e.g., K - Digestive System). This looser metric assesses whether the model situates the diagnosis in the correct broad clinical domain, even when more fine-grained distinctions are missed.

Together, these three hierarchical metrics allow us to quantify both the coding precision and the clinical proximity of the generated outputs. This layered evaluation is particularly important given the granularity of ICD-10-CM.

## 3.5 Knowledge Graph Visualisation

Triples were extracted from the LLM outputs of each prompting framework using regular expressions. Although KGGen, EDC, and Text2KGBench each produced outputs containing entities and relations, their formats differed. Text2KGBench followed a defined triple structure, which was used as an example for clarity. The extracted triples were then used to construct directed graphs in Python using the `NetworkX` library, where nodes represented entities and edges represented relations. These graphs were rendered for inspection, and an example of the extracted triples is shown in Appendix G. Visualising the outputs provided an accessible way to examine the extracted information qualitatively and demonstrate potential for visual representation of text.

# 4 Results

## 4.1 Data Stratification

Figure 8 shows the distribution of the initial category of the ground-truth principal diagnosis codes before and after stratification. In the full dataset (left), the distribution is highly imbalanced, with certain categories (e.g., K, Z, S) dominating the sample, while others occur only rarely. After stratification (right), the relative proportions across categories are more evenly adjusted, ensuring that underrepresented categories are not disproportionately excluded.

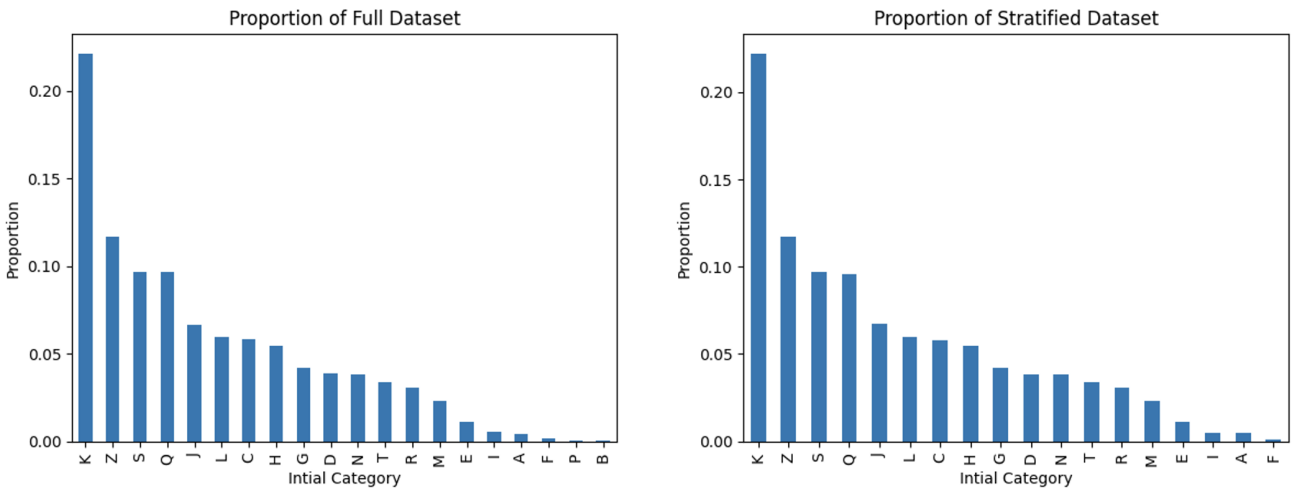


Figure 8: Distribution of the initial category of the ground-truth principal diagnosis in the entire dataset (left) and the stratified dataset (right)

It is also notable that some categories present in the full dataset, such as P and B, are no longer represented in the stratified dataset. This reflects their very low frequency in the original distribution, which made them statistically negligible once the dataset was capped at 1,000 samples. The stratified set provides a balanced and manageable experimental sample while still retaining the major trends of the full dataset.

## 4.2 Baseline Results of Classification

The baseline comparison shows that Llama-3-70B consistently outperforms MedGemma-27B across all levels of correctness. The gap is most pronounced at the category match level (0.46 vs. 0.219), establishing Llama-3-70B as the stronger baseline model for subsequent experiments.

| | **Llama-3-70B** | **MedGemma-27B** |
|---|---|---|
| Exact Match | 0.11 | 0.024 |
| Category Match | 0.46 | 0.219 |
| Initial Category Match | 0.656 | 0.445 |

Table 5: Baseline performance of principal diagnosis using Llama-3-70B and MedGemma-27B as classifiers.

## 4.3 Classification from Constructed Knowledge Graphs

### 4.3.1 Exact Match Metric

When the Llama-3-70B model was employed as the classifier (Table 6), generating the ICD-10-CM principal diagnosis code from knowledge graph input, EDC achieves the highest exact match score (0.105), slightly improving upon KGGen (0.092) and Text2KGBench (0.083). For MedGemma-27B (Table 7), KGGen produces the best result (0.038), though overall performance remains lower than with Llama-3-70B. These results suggest that EDC is better suited to larger general-purpose models, while KGGen aligns more effectively with MedGemma.

| | **Mistral-Small-22B** | **Llama-3-8B** | **Llama-3-70B** | **MedGemma-27B** |
|---|---|---|---|---|
| KGGen | **0.092** | **0.088** | 0.094 | **0.1** |
| EDC | 0.087 | 0.06 | **<u>0.105</u>** | 0.09 |
| Text2KGBench | 0.083 | 0.08 | 0.089 | 0.085 |

Table 6: Exact match accuracies by prompting style using Llama-3-70B as the classifier.

| | Mistral-Small-22B | Llama-3-8B | Llama-3-70B | MedGemma-27B |
|---|---|---|---|---|
| KGGen | **0.031** | **0.031** | **0.038** | **0.036** |
| EDC | 0.028 | 0.02 | 0.033 | 0.033 |
| Text2KGBench | 0.017 | 0.025 | 0.017 | 0.027 |

Table 7: Exact match accuracies by prompting style using MedGemma-27B as the classifier.

#### 4.3.2 Category Match Metric

Using Llama-3-70B (Table 8), EDC again provides the strongest performance (0.448), with KGGen (0.401) and Text2KGBench (0.407) performing similarly. In contrast, MedGemma-27B (Table 9) shows its best results with Text2KGBench (0.214) and EDC (0.253), suggesting that ontology-guided prompting has greater benefit for the domain-specific model.

| | Mistral-Small-22B | Llama-3-8B | Llama-3-70B | MedGemma-27B |
|---|---|---|---|---|
| KGGen | 0.369 | 0.345 | 0.401 | 0.36 |
| EDC | **0.404** | 0.246 | **0.448** | **0.424** |
| Text2KGBench | 0.393 | **0.353** | 0.407 | 0.391 |

Table 8: Category match accuracies by prompting style using Llama-3-70B as the classifier.

| | Mistral-Small-22B | Llama-3-8B | Llama-3-70B | MedGemma-27B |
|---|---|---|---|---|
| KGGen | 0.202 | 0.192 | 0.223 | 0.184 |
| EDC | 0.197 | 0.134 | **0.231** | **0.253** |
| Text2KGBench | **0.214** | **0.204** | 0.169 | 0.212 |

Table 9: Category match accuracies by prompting style using MedGemma-27B as the classifier.

#### 4.3.3 Initial Category Match Metric

At the broader initial category level, all methods achieve higher scores. With Llama-3-70B (Table 10), EDC reaches the best performance (0.657), closely followed by Text2KGBench (0.643). For MedGemma-27B (Table 11), EDC again

achieves the highest score (0.524), followed closely by Text2KGBench (0.507). This suggests that knowledge graph construction improves classification performance for both models when evaluated at the broader, initial-category level.

| | **Mistral-Small-22B** | **Llama-3-8B** | **Llama-3-70B** | **MedGemma-27B** |
|---|---|---|---|---|
| KGGen | 0.581 | 0.598 | 0.638 | 0.565 |
| EDC | 0.608 | 0.429 | **0.657** | **0.628** |
| Text2KGBench | **0.643** | **0.623** | 0.628 | 0.626 |

Table 10: Initial category match accuracies by prompting style using Llama-3-70B as the classifier.

| | **Mistral-Small-22B** | **Llama-3-8B** | **Llama-3-70B** | **MedGemma-27B** |
|---|---|---|---|---|
| KGGen | 0.457 | 0.447 | **0.492** | 0.455 |
| EDC | 0.453 | 0.329 | 0.489 | **0.524** |
| Text2KGBench | **0.507** | **0.497** | 0.441 | 0.501 |

Table 11: Initial category match accuracies by prompting style using MedGemma-27B as the classifier.

## 4.4 Visualised Knowledge Graphs

To illustrate the outputs produced through knowledge graph construction, two representative visualisations generated using the Text2KGBench framework are presented below. These examples, constructed with the Llama-3-70B (Figure 9) and MedGemma-27B (Figure 10) models, highlight differences in graph granularity and term specificity that arise from variations in model scale and domain alignment. Each visualisation depicts entities as nodes and their relationships as directed edges, providing a qualitative view of how the extracted information is structured. While quantitative metrics are reported, these visualisations offer a depiction of the structural variation in knowledge graph outputs observed across models.

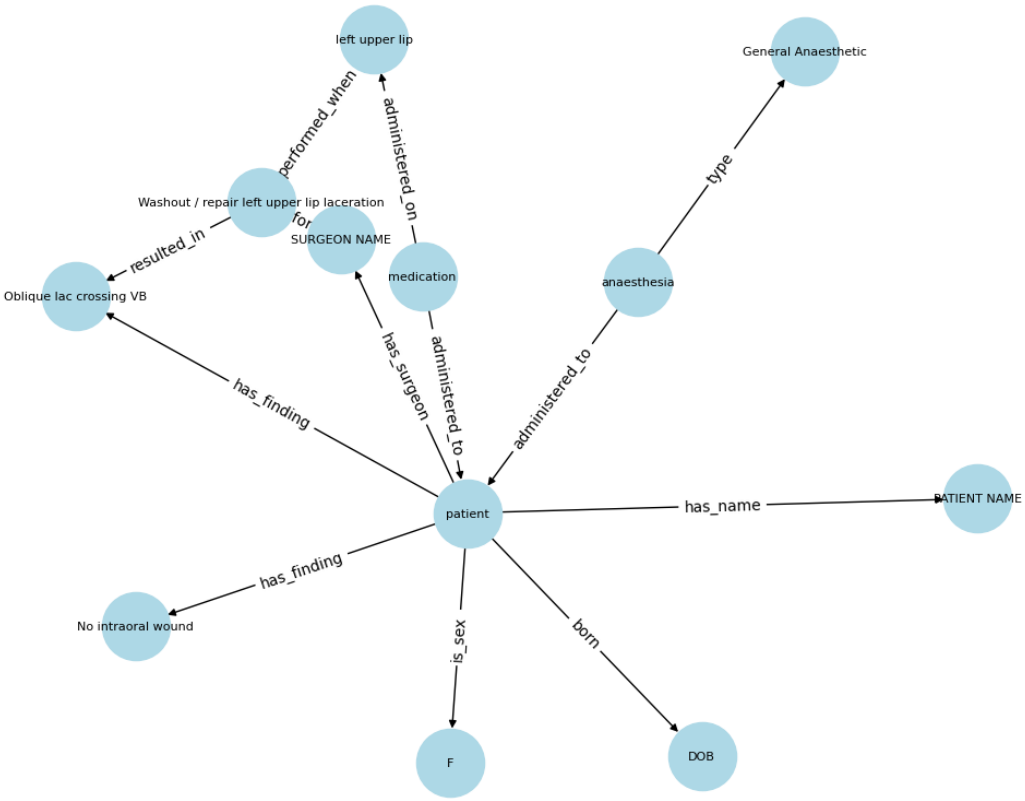


Figure 9: Visualised knowledge graph using Llama-3-70B and Text2KGBench prompt. The ontology-aligned structure presents a more abstract and generalised representation of the clinical text.

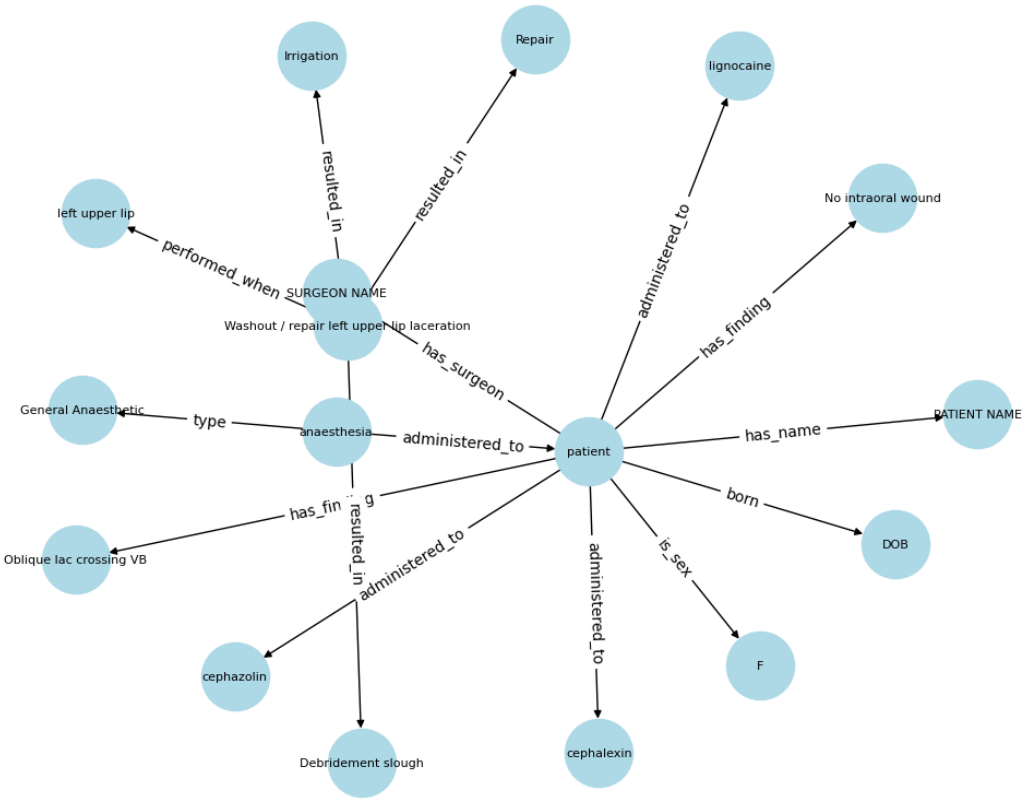


Figure 10: Visualised knowledge graph using MedGemma-27B and Text2KGBench prompt. The medically tuned model incorporates richer terminology and greater domain-specific detail.

# 5 Discussion

## 5.1 Performance Across Levels of Granularity

From our results in Section 4.3, we observe a clear trend where model performance improves as the level of granularity in the ICD-10-CM evaluation is relaxed. In other words, when the classification task is assessed at broader categorical levels, models demonstrate higher agreement with the ground truth compared to the stricter, exact-code evaluations. This pattern reflects the hierarchical nature of the ICD-10-CM coding system, where minor variations in clinical detail or phrasing can lead to different specific codes within the same overarching diagnostic category. Consequently, as the evaluation shifts toward coarser levels of granularity, the models' predictions align more frequently with the correct diagnostic family, even if the precise sub-code differs.

This trend highlights the broader motivation of this research—to examine whether structured representations can offer a more interpretable and standardised way of representing clinical information, rather than necessarily achieving higher raw predictive accuracy. Healthcare documentation varies widely across institutions, and small differences in language or structure can alter code assignment even when the underlying diagnosis is the same. By constructing knowledge graph representations, our research explores an approach aimed at improving semantic clarity and interoperability, ensuring that clinical narratives can be consistently interpreted even when coding outcomes differ at the most specific levels.

### 5.1.1 Performance at Exact Match

The knowledge graphs generated using KGGen's two-step extraction and relation-linking framework achieved the strongest overall performance across most model pairs, with the only exception being the EDC framework when paired with the Llama-3-70B model. As illustrated in Figures 11 and 12, the graphs produced through KGGen and EDC both capture more fine-grained and semantically rich connections between entities compared to simpler extraction baselines. This increased granularity enables the downstream model to reason over a more detailed

and contextually complete representation of each clinical scenario. Consequently, the model is better informed to assign an appropriate and more specific ICD-10-CM principal diagnosis code, rather than defaulting to a higher-level or more general category. In other words, by explicitly modelling the relationships between clinical entities, such as procedure, complication, and anatomical site, these frameworks provide a structured representation that narrows the model's interpretive ambiguity and improves precision in diagnostic classification.

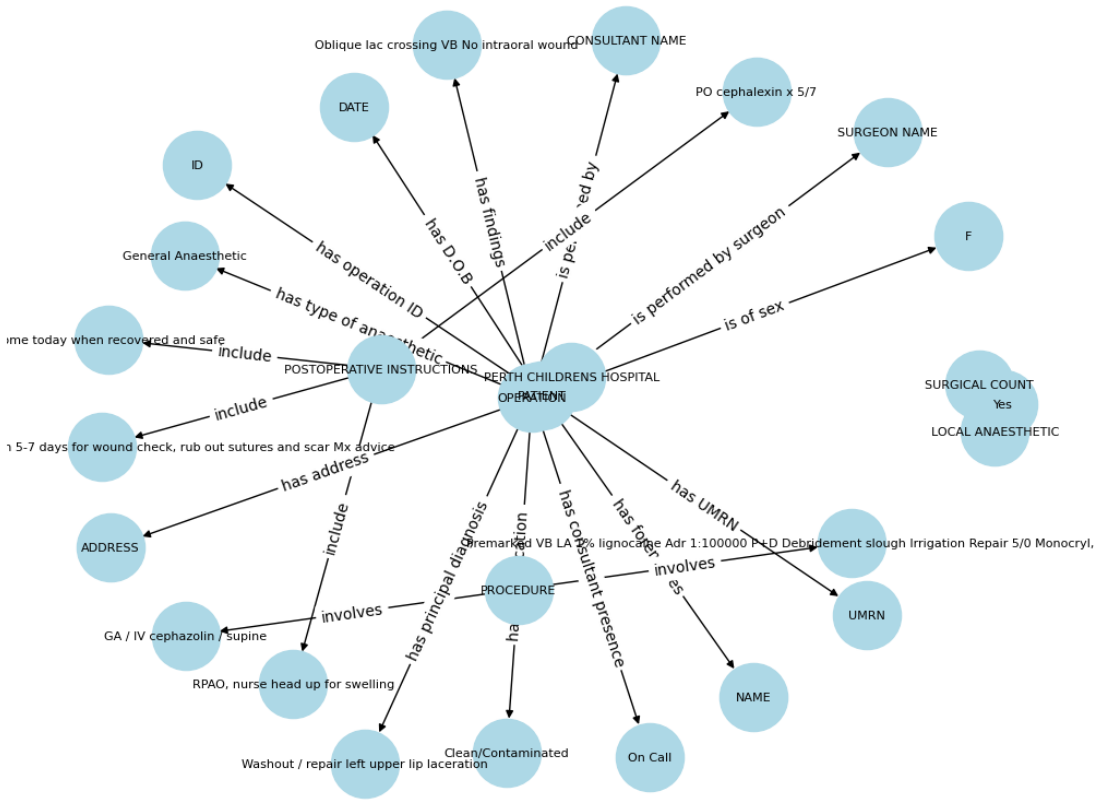


Figure 11: Visualised knowledge graph using Llama-70B and KGGen prompt

However, the same fine-grained nature of KGGen that contributes to its strength in exact code prediction can also introduce challenges under more relaxed evaluation criteria, such as the initial category match (Table 10, 11). This is because KGGen captures a highly detailed network of entities and relations, which the res-

ulting knowledge graph can become too specific relative to the broader diagnostic categories represented at the initial level of ICD-10-CM codes. In these cases, the model is exposed to a larger set of potentially relevant but overlapping signals, making it more difficult to isolate the single principal diagnosis that aligns with the higher-level ICD category. As a result, although KGGen improves the model's capacity for precise classification, it may inadvertently reduce performance on metrics that align with more general agreement. Thereby, reflecting the trade-off between semantic depth and categorical generality in knowledge graph-informed prediction.

Figure 12: Visualised knowledge graph using Llama-70B and EDC prompt

### 5.1.2 Performance at Broader Category Levels

When evaluated at broader categorical levels, such as the category match and initial category match, the relative performance among frameworks shifts considerably. While KGGen continues to demonstrate consistent accuracy, the Text2KGBench framework exhibits a substantial improvement in this setting compared to the exact code evaluation. This reversal in trend reflects the nature of the ontology and extraction design used within Text2KGBench.

As illustrated in Figure 5, Text2KGBench structures its knowledge representation around high-level clinical entities such as *patient*, *procedure*, *findings*, and *anatomy*, connected through direct and semantically broad relations like *performed_on*, *resulted_in*, and *has_surgeon*. This ontology captures essential diagnostic and procedural context without extending into the highly detailed, nested relationships found in frameworks such as KGGen. In effect, it abstracts clinical text into a generalised conceptual snapshot of the operation report, prioritising interpretability and top-level associations over granular specificity.

This abstraction becomes advantageous when the ICD-10-CM classification task is assessed at coarser levels of granularity. This is because the knowledge graphs produced by Text2KGBench highlight broad diagnostic cues, rather than detailed procedural or complication-level information (Figure 9). The model aligns more effectively with the overarching diagnostic categories that define the initial characters of ICD codes. The result is a higher rate of partial or categorical matches, as the extracted representations map cleanly to the conceptual boundaries of ICD-10-CM system.

However, the same simplification that supports this broader alignment also limits performance under exact-match evaluation. Without encoding fine-grained relationships (e.g., between complications, procedure steps, or secondary findings), the model lacks the contextual precision needed to differentiate between specific sub-codes within the same category. Consequently, while Text2KGBench improves robustness and generality at broader classification levels, it does so at the expense of precision, illustrating the inverse relationship between information granularity and categorical performance observed across all frameworks.

### 5.1.3 Implications for Structured Representation

Taken together, these patterns demonstrate that the level of structural detail embedded in a knowledge graph directly influences how effectively an LLM can align clinical text with hierarchical diagnostic taxonomies such as ICD-10-CM. Frameworks like KGGen excel when fine-grained clinical relationships are required, offering a precise and semantically dense representation of operative context. In contrast, Text2KGBench performs more robustly when the task rewards broader, concept-level associations, where the abstraction of complex narratives into core entities and relations provides a more interpretable foundation for classification.

Rather than viewing these differences purely in terms of predictive performance, they should be understood through the lens of the research motivation. The primary objective of our research is not to demonstrate that structured representations outperform free-text classification, but to explore how they contribute to standardisation and semantic clarity in clinical documentation. Variability in how surgical information is written and interpreted across practitioners and institutions remains a major barrier to consistent ICD-10-CM coding and data interoperability. By constructing knowledge graph representations, this research proposes a framework through which unstructured narratives can be transformed into coherent, reusable structures that support interoperability, reproducibility, and transparent reasoning. Ultimately, these findings reinforce the central motivation of this work, which is to develop methods that advance the interpretability and interoperability of clinical information, even when immediate accuracy gains are modest.

## 5.2 Impact of Using a Medically Tuned Model for Knowledge Graph Construction

Beyond the influence of the extraction framework itself, the underlying language model used for knowledge graph construction also plays a critical role in shaping the semantic depth and domain specificity of the resulting graphs. When comparing graphs generated using the general-purpose Llama-3-70B model with those constructed using the medically tuned MedGemma-27B model, we observe notable differences in both vocabulary richness and the precision of clinical entity linking.

As illustrated in Figure 10, the knowledge graph generated using the MedGemma-27B model in conjunction with the Text2KGBench framework demonstrates a markedly greater inclusion of medical-specific terminology and entities. For instance, it identifies pharmacological terms such as *lignocaine*, alongside procedural entities that are not seen in Figure 9. This expanded terminology coverage enables the graph to represent a more contextually grounded and clinically faithful view of the operation report, enhancing its interpretive value for downstream diagnostic inference.

These improvements are evident when comparing the outputs of frameworks like KGGen and Text2KGBench across model types. While Llama-3-70B tends to produce higher-level or generic descriptors (Figure 11), the MedGemma-27B model refines these into more specialised instances, specifying agents, medication, or procedural details. Consequently, the resulting graphs exhibit denser and more clinically meaningful connections, which better align with structured vocabularies such as those underlying the ICD-10-CM classification system.

These observations reinforce the broader motivation for employing domain-specific language models in clinical contexts. In healthcare, the terminology used within documentation is often highly specialised, context-dependent, and inconsistent across institutions. Medically tuned models such as MedGemma provide a mechanism for aligning extracted information with standardised vocabularies and ontologies. Thereby producing knowledge graphs that more faithfully capture the semantic content of unstructured clinical text. This alignment is crucial for transforming free-form narratives into structured, machine-interpretable representations that can integrate with existing health information systems.

## 5.3 Model Scale, Ontology Structure, and Prompting Complexity

An additional dimension influencing performance across frameworks is the scale of the language model employed during knowledge graph construction and classification. As shown in Tables 10 and 11, smaller models such as Mistral-Small-22B and Llama-3-8B exhibit greater relative gains when guided by structured

frameworks like Text2KGBench. In contrast, larger models such as Llama-3-70B and MedGemma-27B maintain strong performance even under less constrained prompting strategies, suggesting that their internal language representations already capture a substantial amount of relational structure.

The design of the Text2KGBench ontology provides a critical advantage for smaller parameter models. By enforcing a simplified, schema-aligned representation of the clinical text, consisting of broad yet semantically interpretable entities (e.g., *procedure*, *findings*, *anatomy*) and direct relations (e.g., *performed_on*, *resulted_in*), the framework effectively reduces the reasoning difficulty on the model. This structure narrows the interpretive search space, guiding smaller models toward relevant contextual associations that they might otherwise overlook. The resulting improvement in alignment between extracted triples and ICD-10-CM diagnostic categories is evident in their consistently higher initial category match scores under Text2KGBench prompting.

Conversely, as model capacity increases, the need for rigid ontological scaffolding diminishes. Larger models, such as Llama-3-70B, demonstrate the ability to generalise across clinical relations and lexical variations without extensive structural constraints. In such cases, a lightweight few-shot prompting approach can be sufficient to achieve comparable or even superior results. These models leverage their broader contextual understanding to infer implicit relations between entities, effectively internalising much of the reasoning that smaller models require explicit structured guidance to perform.

However, these benefits must be weighed against the substantial computational costs associated with deploying large-scale models in healthcare environments. Models such as Llama-3-70B or MedGemma-27B demand significant GPU resources, memory capacity, and inference time, which may exceed the infrastructure available to many health institutions. In such contexts, Text2KGBench provides a highly practical alternative. Its ontology-guided prompting allows smaller and more cost-efficient models to achieve strong alignment without requiring high-end hardware or prolonged inference runtimes. In contrast, institutions equipped with sufficient computational resources can leverage frameworks like EDC in combina-

tion with large models to achieve high-quality extractions without the need for rigid ontological structures. EDC's single-prompt extraction process, when paired with models of sufficient scale, achieves a favourable balance between efficiency and semantic richness. Ultimately, the choice of framework and prompting strategy should align with both the model capacity and the operational capabilities of the implementing organisation. This is where Text2KGBench offers accessibility and stability for resource-limited settings, and EDC serves as an effective, lightweight alternative for institutions with access to large, domain-tuned models.

From a broader perspective, these findings highlight the importance of aligning technical design choices with the real-world constraints and objectives of healthcare environments. The motivation behind this work extends beyond the comparison of raw model performance and focuses on developing practical methods for representing and reasoning over clinical information that are both interpretable and operationally feasible. Large models provide rich contextual reasoning but require computational resources that may be inaccessible to many hospitals. In contrast, smaller, ontology-guided approaches such as Text2KGBench offer a scalable pathway toward interoperability and standardisation, enabling institutions with limited infrastructure to adopt structured, semantically coherent representations of their clinical data.

## 5.4 Limitations

One limitation of our implementation concerns the post-processing strategy used to extract ICD-10-CM codes from the large language model outputs. In cases where a model returned multiple candidate codes, only the first valid code detected by the regular expression was retained for evaluation. This strict "first-code only" policy ensured consistency across models and prevented artificially inflated scores resulting from multiple guesses. However, it also introduced the risk of underestimating model performance, particularly in instances where the correct principal diagnosis appeared later in the response sequence.

This behaviour is illustrated in Appendix F, where model outputs contain multiple ICD-10-CM codes proposed. While such responses demonstrate that the model had

correctly identified the appropriate diagnostic domain, the conservative extraction policy focused on only the first occurrence, discarding potentially valid codes that followed. As a result, models that exhibited hedging behaviour by producing a list of plausible candidates rather than a single definitive prediction may have been disproportionately penalised under the adopted evaluation framework.

The decision to retain only the first valid code was a deliberate trade-off between methodological simplicity and completeness. This approach prioritised consistency and replicability across models, ensuring that all evaluations were performed under the same strict criteria. Nonetheless, examining model outputs through a top-k evaluation lens could provide a more balanced perspective on predictive behaviour, allowing a distinction between complete misclassification and partial correctness.

# 6 Conclusions

This research investigated the use of large language models for automated knowledge graph construction from unstructured surgical operation reports, with the goal of improving the interpretability and structured representation of clinical information. By implementing and comparing three prompting frameworks, KG-Gen, EDC, and Text2KGBench, across multiple model scales and domains, the study evaluated how different extraction strategies influence both the quality of generated knowledge graphs and their utility for ICD-10-CM principal diagnosis classification. While the incorporation of structured representations did not lead to substantial gains in exact match accuracy relative to plaintext classification, improvements were observed at broader categorical levels of evaluation. These results reflect the hierarchical nature of ICD-10-CM coding and suggest that knowledge graph representations can enhance semantic alignment and category-level understanding, even when exact diagnostic precision remains challenging.

Beyond classification performance, the study highlights the broader value of constructing interpretable and standardised representations of clinical text. The proposed lightweight ontology, centred on operative relationships between patient, procedure, anatomy, and findings, establishes a foundation for developing patient-centric summaries that condense complex surgical narratives into structured, clinically coherent views of each hospital encounter. Capturing who performed the procedure, what was done, where it occurred, and what the findings were it supports automatic generation of concise yet contextually complete accounts of care. Compared with comprehensive clinical terminologies such as SNOMED-CT, this schema remains intentionally lightweight and focused, allowing it to function efficiently within surgical documentation while retaining the flexibility to align with broader ontological frameworks in future work. This approach moves toward representations that are not only machine-readable but also intuitively interpretable by clinicians, facilitating clearer communication and a more transparent audit of clinical decisions.

Overall, this work contributes to ongoing efforts in clinical natural language processing by demonstrating that large language models, when guided by appropriate

prompting strategies, can extract structured and clinically meaningful representations from complex free-text data. The findings underscore the trade-off between granularity, interpretability, and computational feasibility in healthcare applications, pointing toward the need for scalable frameworks that balance precision with accessibility.

## 6.1 Future Work

While this study has focused primarily on extracting structured information from post-operative reports, future research should extend beyond this single time point to capture a more comprehensive and temporally aware representation of patient care. As illustrated in Figure 13, this progression involves linking information across the full peri-operative timeline, including admissions, operations, progress notes, and discharge summaries, to develop a longitudinal understanding of each patient's clinical journey.

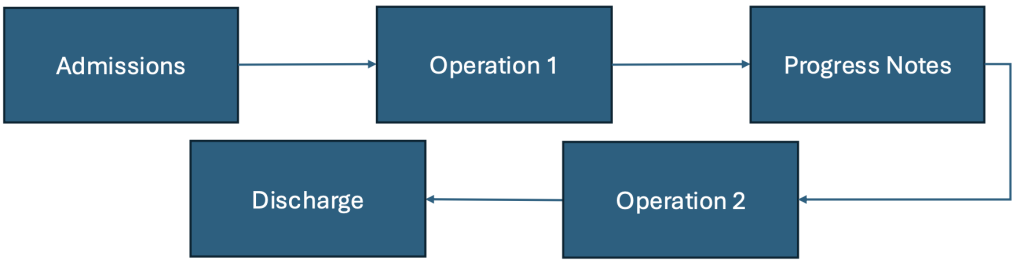


Figure 13: Example flow of a patient's hospital visit

### 6.1.1 Temporal and Longitudinal Integration

The methodology presented in this work focuses on static extraction from individual operation reports. However, patient care is inherently dynamic and evolves through multiple clinical encounters, interventions, and revisions. Future work should therefore aim to incorporate a complete timeline of reasoning into the knowledge graph construction process, enabling the representation of how diagnoses, procedures, and administered treatments unfold chronologically.

By aligning entities and relations across time, it becomes possible to model the patient's trajectory from admission to discharge, capturing the sequence and interdependence of clinical events. This temporal structure would enrich each knowledge graph, allowing models to reason not only over what occurred but also when and in what context. In turn, this could further facilitate longitudinal analyses such as identifying postoperative complications, monitoring recovery patterns, or correlating interventions with outcomes. This temporal linking aligns with the broader motivation of improving the interpretability and continuity of clinical records, transforming static documentation into a coherent, patient-centric narrative.

### 6.1.2 Integration with Controlled Vocabularies and External Medical Knowledge

A complementary direction involves grounding extracted knowledge in external medical resources and controlled vocabularies to improve semantic consistency and interoperability. Recent advances, such as the MedGraphRAG framework [8], demonstrate how retrieval-augmented generation (RAG) can connect user-provided documents with verified biomedical databases and ontologies. Adapting similar strategies within this research context would allow extracted entities and relations to be cross-referenced with credible sources such as SNOMED-CT [19], ensuring that the generated graphs are both clinically accurate and semantically standardised.

This grounding would enable knowledge graphs to act as interoperable bridges between institutional datasets, supporting the harmonisation of terminology and meaning across healthcare systems. Moreover, linking extracted representations to trusted medical vocabularies could enhance explainability and reliability in LLM outputs, as each node and relation in the graph could be explicitly traced to a concept definition.

### 6.1.3 Towards a Unified and Interoperable Representation of Perioperative Care

The integration of temporal reasoning and controlled medical knowledge offers a pathway toward developing a unified representation of perioperative care. By connecting clinical data across time and aligning it with external biomedical resources, future frameworks could produce knowledge graphs that are both contextually rich and universally interpretable. This evolution would extend the practical utility of the system beyond diagnostic classification, supporting tasks such as patient history summarisation, clinical question answering, and retrieval-augmented decision support.

Ultimately, the long-term goal is to move toward a machine-interpretable model of healthcare data that captures not only the factual content of individual reports but also the evolving structure of patient care across encounters. Achieving this would represent a significant step toward interoperable, longitudinal electronic health records capable of supporting clinical reasoning, institutional analytics, and cross-system data integration.

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# 8 Appendices

# A Declaration of AI Use

Large Language Models (LLMs) were used in this work to assist with language refinement, grammar correction, and code optimisation. Specifically, LLMs supported the editing of written material to improve readability and grammatical accuracy. For coding-related components, I provided the core structure, logic, and implementation, while LLMs were used to enhance efficiency, identify potential errors, and ensure consistency across scripts. All research concepts, methodological design, analysis, and interpretation remain entirely my own work.

# B KGGen Prompt

**Prompt for extracting entities:** `Extract key entities from the given text. Extracted entities are nouns, verbs, or adjectives, particularly regarding sentiment. This is for an extraction task, please be thorough and accurate to the reference text.`

**Prompt for extracting relations:** `Extract subject-predicate-object triples from the assistant message. A predicate (1-3 words) defines the relationship between the subject and object. Relationship may be fact or sentiment based on assistant's message. Subject and object are entities. Entities provided are from the assistant message and prior conversation history, though you may not need all of them. This is for an extraction task, please be thorough, accurate, and faithful to the reference text.`

# C EDC Prompt

Given a piece of text, extract relational triplets in the form of [Subject, Relation, Object] from it.
Here are some examples:
Example 1:
Text: The 17068.8 millimeter long ALCO RS-3 has a diesel-electric transmission.
Triplets: [['ALCO RS-3', 'powerType', 'Diesel-electric transmission'], ['ALCO RS-3', 'length', '17068.8 (millimetres)']] ...

Now please extract triplets from the following text: Alan Shepard was born on Nov 18, 1923 and selected by NASA in 1959. He was a member of the Apollo 14 crew.

# D Text2KGBench Prompt

*Given the following ontology, examples and sentences, please extract the triples from the sentence according to the relations in the ontology. In the output, only include the triples in the given output format.*

***CONTEXT:***

**Ontology Concepts:** *human, city, country, film, film genre, film production company, film award, award, written work, film character, film organization*

**Ontology Relations:** cast_member(film,human), director (film,human), screenwriter (film,human), producer(film,human), genre(film,genre), based_on(film,written work), award_received (film,award), production_company(film,film production company), country_of_origin(film,country), publication_date (film,date), characters(film,film character), narrative_location(film,city), filming_location(film,city), main_subject(film,thing), nominated_for(film,award), cost(film,number)

***Example Sentence****: The Lion King is a animated musical drama film about a lion cub who is to succeed his father and it was directed by Roger Allers and Rob Minkoff (in their feature directorial debuts), produced by Don Hahn.*

***Example Output:*** director(Lion King, Roger Allers)
director(Lion King, Rob Minkoff)
producer(Lion King, Don Hahn)

***Test Sentence:*** Birds Anonymous is a 1957 Warner Bros. Merrie Melodies animated short, directed by Friz Freleng and written by Warren Foster.

***Test Output:***

# E Example Clinical Text

8 year old male
Notes: 1. Emergency Department medical note 1/1/25 8 year old boy NKDA No medications Last meal 8:00am
Developed right sided testicular pain 1/7 ago. Did not tell anybody as did not want to worry anyone. Ate a sandwich on the way to hospital at 0730. USS showing avascular right testes - referred to surgeons at PCH
O/E Comfortable Not examined further
Plan Admit to hospital NBM Surgeons informed
2. Surgical note 1/1/25 Paediatric Surgery Admission 8M with right testicular torsion on ultrasound HPC One day history of right sided scrotal pain Initially had vomiting but none in past 6h No fevers No urinary symptoms No abdominal pain
B/g Nil previous operations Well
O/E Obs WNL Comfortable Swollen right hemiscrotrum
d/w Dr French
Plan NBM B&C right scrotal exploration +/- right orchidectomy
3. Consent note 1/1/25 Scrotal exploration +/- right testis +/- right orchidectomy
4. Nursing note 1/1/25 IVC inserted Patient comfortable No concerns ATOR
5. Operation report Operation and principal diagnosis: Right sctrol exploration and orchidectomy Indication 8M with 1 day history of pain
Findings Oedema and swelling of right hemiscrotum 720 degree torsion of testis with non viable ischaemic testis, black epididymis Decision made not for orchidopexy given swelling and inflammation
Procedure details Team time out IV cefazolin Betadine prep and drape Median raphe incision Testis delivered Detorted Warm pack for 15 minutes, no improvement Orchidectomy performed Divided with diathermy Haemostasis Dartos closed with 3-0 vicryl Skin closed with 4-0 vicryl rapide Steristrips + opsite spray
Postoperative instructions Diet as tolerated No sport for 2 weeks Follow up with Dr French regarding management Home tomorrow if well
Discharge summary 2/1/25 [...] Advice to parents We wish you all the best on discharge We advise that patient stay off school for 2-3 days to guide return to school based on comfort levels and symptoms Patient should not swim for 2 weeks We will see you in clinic in 2-3 weeks time.
Clinic letter 21/1/25

# F Example poor outputs for ICD Classification from LLMs

**Llama-3.3-70B**
"M86.671, F10.929, T45.515A, Y83.318, Z85.49, Z98.89"
**MedGemma**
"“‘ Z00.00, Z00.01, Z00.02 “‘"

# G Example Extracted Triples

[has_name(patient, PATIENT NAME), is_sex(patient, F), born(patient, DOB), has_surgeon(patient, SURGEON NAME), performs(SURGEON NAME, Washout / repair left upper lip laceration), performed_when(Washout / repair left upper lip laceration, left upper lip), has_finding(patient, Oblique lac crossing VB), has_finding(patient, No intraoral wound), administered_to(patient, cephazolin), administered_to(patient, lignocaine), administered_to(patient, cephalexin), type(anaesthesia, General Anaesthetic), administered_to(anaesthesia, patient), resulted_in(Washout / repair left upper lip laceration, Debridement slough), resulted_in(Washout / repair left upper lip laceration, Irrigation), resulted_in(Washout / repair left upper lip laceration, Repair)]

# Original Research Proposal

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Degree: Bachelor of Advanced Computer Science (Honours)

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## 1 Background

The experience of undergoing surgery can be overwhelming for both children and their caregivers. The perioperative process often leads to behavioral and psychological changes in the child and significant emotional distress for the caregiver [1]. Complications that arise from surgery can be handled by medical professionals if they are still in pediatric care. However, once they have been discharged, monitoring their health falls to carers. This can be a struggle for those not from a healthcare background. Previous methods to monitor the recovery of postoperative children were over the telephone, whereas recent studies have found conversational SMS to be more feasible and more preferred by caregivers [2].

To advance this, generative artificial intelligence (AI) has been applied to generate responses from user input [3]. The benefits of this application are that it can save costly time for doctors to skip reading through records to understand a patient's medical history to provide a healthcare solution. Working with AI, this process can be sped up with clinicians overseeing the generated responses to an SMS message before they are relayed to patients and carers.

### 1.1 Previous Works

Text2KGBench establishes a benchmarking framework designed to evaluate the effectiveness of text-to-knowledge graph generation models [4]. By incorporating datasets and structured evaluation metrics, Text2KGBench provides a standard-

ised platform for assessing model performance across accuracy, coherence, and clinical relevance. This framework not only enhances the development of reliable knowledge graphs but also facilitates the investigation of generative AI's potential in structuring complex medical knowledge graphs.

MedGraphRAG introduced a three-level hierarchical graph structure that links clinical notes, a medical dictionary, and medical papers [3]. This integration enables referenced responses, enhancing both the reliability and trustworthiness of LLMs in medical applications. This structure paves the way for exploring more specialised generative capabilities within targeted medical domains, driving advancements in clinical support.

# 2 Project Aim

This project aims to enhance knowledge graph extraction from patient documents. The extracted knowledge graphs will be leveraged and integrated into an LLM to predict post-operation complications that arise from surgery.

1. Investigate the sets of entity and relationship types that define the structure of a knowledge graph in the medical context accurately
2. Experimenting to find the most effective extraction methods of patient data to form medical knowledge graphs and integrate them with LLMs
3. Evaluate the effectiveness of using a knowledge graph in LLM for prediction of perioperative complications.

# 3 Methods

We have access to several pediatric documentation datasets containing detailed records of patients who have undergone surgery. Our goal is to construct a comprehensive patient knowledge graph that integrates these datasets, enabling us to enhance various downstream tasks, including risk prediction, clinical decision support, and generative modeling for personalised healthcare insights. Previous investigations include MedGraphRAG, which is relevant to the medical domain, and MaintIE, although not specific to the medical field, it is relevant for understanding entity and relationship extraction.

MaintIE is a research project that developed an annotation schema from maintenance texts for entity recognition and relation extraction. These consisted of 224 entities and 6 relations all crafted to suit the context of maintenance [5]. This application of a personalised information extraction applies to the medical field

due to the similarities in the presentation of clinical notes and maintenance texts, which tend to be short and abbreviated sentences related to one another.

"CBA321 BROK CLAMP ON CROWD CYL HOSE TBC"
"USS showing avascular right testes - referred to surgeons at PCH"

Figure 1: **Top** — Example maintenance short text [5]
**Bottom** — Example description in clinical notes

SNOMED-CT is a set of standards implemented in eighty different countries including Australia. These standards are defined for electronic exchange of clinical information with terminology speeding the process of recording data for clinicians. Developing entity and relationship types based on SNOMED-CT terminology is important to continue the standard of information representation in the field. It is also important because the use of LLM using the knowledge graph should present its response in a manner that a clinician can understand instantly and not have to spend precious time on searching up synonymous definitions.

Another set of medical classifications are the tenth revision of the International Classification of Diseases (ICD-10). We can leverage ICD-10 codes as an ontology structure, given that we have access to ground truth ICD-10 codes for patient admissions. By structuring patient data within this ontology, we can enhance the representation of medical concepts, enabling more effective entity recognition, relationship extraction. Implementing two separate medical ontologies, SNOMED-CT and ICD-10, we can also compare the accuracy of extracting correct information and their respective representations.

Structuring the ontologies is important to guide LLMs to extract the knowledge from unstructured text [4]. This architecture can then be experimented with many existing LLMs including medically developed LLMs. These LLMs have already been trained with clinical notes and so serve as a starting point to investigate knowledge graph construction with medical pre-training [6].

The consistency of LLMs to construct knowledge graphs whilst sticking to the provided entity and relationship types. This statistic can be kept as labels defined in the structure compared to those the LLM created. This consistency score is important, however, the accuracy of the label the LLM chooses is equally important. The accuracy investigates the logical representation of data within the clinical notes. This is quantifiable with the aid of experts within the field to assess the conventions the LLM had chosen in constructing the knowledge graphs.

After extracting the knowledge graph, implementing it into the LLM is the next step. By leveraging pre-existing LLMs that have a foundational training in the

medical domain [6, 7], they can then be finetuned with the knowledge graphs extracted. This can be done by splitting the data into training and testing sets and analysing the predictions made on the testing sets as a key indicator of performance. To evaluate the LLM's generative/predictive ability, comparing it to other existing benchmarks will be critical in determining its capabilities [3, 8] and the ability to correctly stick to the ontological structure. Identifying the improvements and weaknesses that the model has is important to suggest further research in the area, and to build confidence in predictive usage.

# 4 Resources

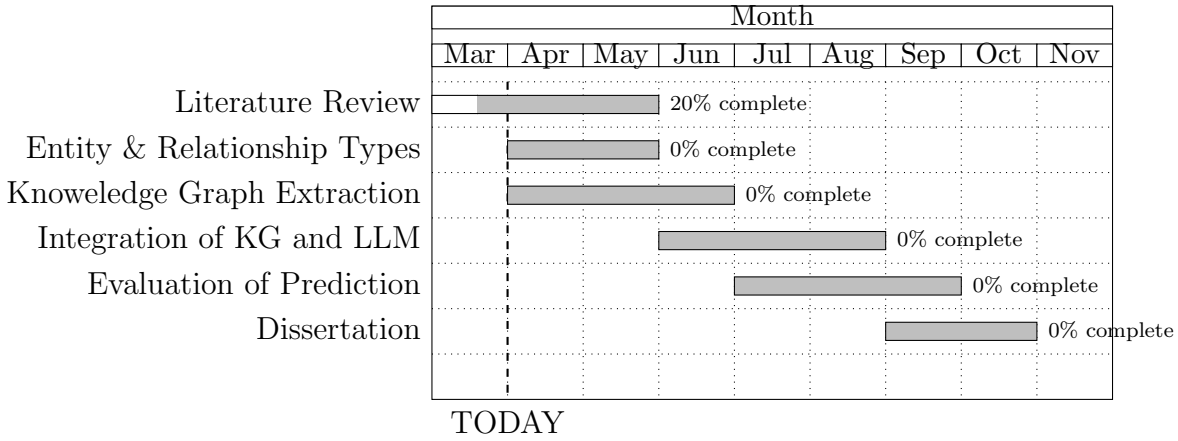
The clinical data this project investigates is from the Anesthesia and Pain Department of Perth Children's Hospital (PCH). Gaining access to the data will take at least a month, however, that process has already started.

Fast-processing machines are readily available. There is access to PCH's in-house computer, MERLIN, which is capable of training LLM models. Alongside MERLIN, Setonix is housed at the Pawsey Research Center, which is the world's fourth greenest and most powerful supercomputer in the southern hemisphere.

Trialing with different LLMs may require access to API keys to utilise them. Some are free, such as Google's Gemini, whereas OpenAI's LLMs require a subscription.

# 5 Project Plan

**Project Timeline**



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